

2 Willow Park
Shantalla
Galway
30th June 2021

Dr Enda Barrett
School of Computer Science
National University of Ireland, Galway

Re: Application for the Higher Diploma in Applied Science (Software Design and Development, Industry Stream)

Dear Dr Barrett,

I have recently completed my final year of a BA in Mathematics & Education in NUI Galway and am excited to be applying for a place in the Higher Diploma in an effort to advance my computer science skills. This course stands out to me as being an attractive combination of lectures, lab work and transition into industry placement where theory can be applied.

I have gained transferable problem-solving skills throughout the various mathematics modules completed as part of my degree and have made an effort over the past year to prepare for the Higher Diploma by completing computer science certificates and working as a computer science tutor to post-primary students. My previous experience on teaching placement, as well as working in customer service, have helped me develop strong communication, time management and writing skills that will help me meet the demands of the industry placement.

I am hopeful to be returning to on-campus teaching and feel that my amicable and positive personality would be an asset to my peers. I aim to achieve as highly in the Higher Diploma as I have in my undergraduate degree, combining a solid foundation in software development with my knowledge of mathematics and education to study further and progress through the Conversion Programme Pathway.

Yours sincerely,



Cian Mac Liatháin

Cian Mac Liatháin

Address: 2 Willow Park, Shantalla, Galway | Email: c.maclithain2@nuigalway.ie | GitHub: github.com/foghlaimoir | Phone: 089 484 6527

EDUCATION

National University of Ireland, Galway

BA in Mathematics & Education

Galway, Ireland

09/2013 - 08/2021

Classification: First-Class Honours (predicted)

- Developed Python skills in modules like Cryptography, Networks, Modelling 2, and Non-Linear Systems, R skills in Applied Statistics 1 and 2 and MATLAB skills in Numerical Analysis 2.
- Received a standing ovation after delivering a 3-day workshop titled 'Creative Coding for Maths Makers' to Transition Year students, students were taught how mathematics and computer programming are related using MineCraft and Java.
- School of Mathematics' Stokes Modelling Workshop 2021: member of the winning team where Python and C# were used to investigate the accuracy of the game engine Unity in carrying out scientific experiments.
- Received University Scholar 2019-2020 Award for achieving 68.17% overall in 3rd Year results.

Diploma in Irish

09/2019 - 08/2021

Classification: First-Class Honours (predicted)

- Completed modules involving the study of grammar, creative writing and oral presentations to a CEFR Level of C1.
- Researched the sociological dynamics of Irish as a minority language in Northern Ireland.

Coláiste na Coiribe

Leaving Certificate

Galway, Ireland

09/2005 - 09/2011

- Completed all schooling through the Irish language achieving 500 points out of a maximum of 600.

WORK EXPERIENCE

Corrib Computer Training

Computer Science Tutor

Galway, Ireland

01/2021 - Present

- Led online workshops using Roblox Studio, Lua and Python to teach 3D Modelling and programming to children aged 12-16 in association with Cruinniú na nÓg and Dyspraxia Ireland.
- Organised and taught computer science classes in post-primary schools teaching Microsoft MakeCode, Python and JavaScript to students aged 14-16.
- Received the ALIVE Presidential Award for volunteering toward the UN Sustainable Development Goal of Quality Education.
- Began as a volunteering role, but earned a paid position based on rapport with students and ability to explain core programming concepts.

Galway Community College

Student Teacher (Mathematics)

Galway, Ireland

10/2019 - 01/2021

- Planned and taught mathematics lessons to 1st Year (Ordinary), 3rd Year (Ordinary) and 5th Year (Foundation) groups. Created student resources, prepared detailed lesson plans in line with the current curriculum and kept daily and weekly reflections.
- Carried out an Action Research Project investigating how a classroom response system can be used to promote a constructivist environment and improve learning by allowing for the gathering and analysis of students' answers.
- Worked closely with my co-operating teacher to improve the wellbeing of students with behavioural problems and reduce the risk of early school leaving.
- Volunteered time for After School Homework Club to help Leaving Certificate students from low socioeconomic backgrounds.

Apple

AppleCare AHA iOS/Mac+ Advisor T1

Galway, Ireland

06/2014 - 06/2016

- Assisted customers with issues involving iOS/macOS devices and Apple IDs. Developed expert troubleshooting skills and a strong understanding of operating systems, hardware and professional knowledgebases.
- Received two AppleCare Excellence awards for excellent metrics with 95% Customer Satisfaction, 86% Issue Resolution, 10% Tier 2 Escalation Rate and 14-minute Call Handling Time on average.
- Won the Customer Satisfaction Elite Award for ranking among the top 16 advisors worldwide based on call metrics and positive customer surveys.

COMPUTER SCIENCE PROJECTS

Superhero Search (Flask, HTML/CSS/JavaScript, SQLite)

A website that allows users to search Marvel characters and find the comics they've appeared in.

Numerical Matrix Processor (Java)

Uses a scanner, switch statement, arrays and various methods to perform operations on a matrix input by the user. Allows for addition, multiplication, transposition and the calculation of the determinant or inverse.

Tic-Tac-Toe (Java, JavaFX)

Developed a simple two-player X's and O's game using boolean variables and JavaFX buttons.

HashMap (Java)

Created a HashMap class with add(), remove() and get() functionality using ArrayLists.

COMPUTER SCIENCE CERTIFICATES

- Oracle's Java Foundations Certified Junior Associate - Java
- University of Helsinki's Java Programming 1 and 2 - Java and JavaFX
- Harvard's CS50: Introduction to Computer Science - C, Python, HTML/CSS/JavaScript, and SQL

SKILLS PROFILE

Independent Learning

Determined and self-motivated work ethic, evidenced by successfully completing projects, certificates and the Irish diploma alongside my final year studies. Quickly adapt to new situations and responsibilities, with a strong belief in lifelong learning.

Problem Solving

Experience troubleshooting for Apple, dealing with daily challenges in the classroom, and developing mathematical maturity throughout my degree has led to robust problem-solving skills transferable to software development.

Teamwork

Appreciate the value of working within a team, based on the findings of educational research as well as my own experiences. Consistently make an effort to promote a positive atmosphere where people feel comfortable sharing their perspectives and enjoy working with and learning from my peers.

INTERESTS

Irish

Passionate about the Irish language, make an effort to speak it with friends, read Irish poetry and attend a weekly ciorcal comhrá to keep it in practice. Enjoy following the work of Dr Kevin Scannell, who develops Irish-language software and researches the challenges involved with applying Natural Language Processing techniques to minority languages like Irish.

Reading

Big fan of the fantasy book genre, with the late Robert Jordan's The Wheel of Time series being an all-time favourite. Also enjoy reading various comic books, following particular writers or artists and comparing their interpretations of different characters.

REFEREES

Kathy O'Sullivan
Lecturer in Mathematics Education
School of Education
National University of Ireland, Galway
091 492 195

Evelyn Jackson
School Placement Co-ordinator
Galway Community College
Wellpark Road, Galway
091 755 464

Written Reference 1



Dear Sir/Madam,

Cian MacLiatháin is a final year student on the Bachelor of Mathematics and Education programme in NUI Galway. In the time that I have known Cian, he has proved to be a very conscientious and hardworking student. I have watched Cian grow into a passionate, knowledgeable and caring teacher who has demonstrated a high level of academic achievement. Cian's teaching abilities combined with his strong teamwork skills, interpersonal and communication skills and work ethic have demonstrated his capacity to be an excellent student who progressed to become an excellent teacher.

Cian's final year research project focused on the use of technology in the mathematics classroom to support constructivist teaching and learning. This research project explicitly shows Cian's ability to excel in his academic capabilities. Cian demonstrated a high level of statistical analysis, some of which in my experience, I have only seen in PhD research.

I have every confidence in Cian's academic, professional and personal qualities and abilities and have no doubt he would be an asset to any programme.

Kind Regards,

A handwritten signature in blue ink that reads 'Kathy O'Sullivan' followed by a horizontal line.

Kathy O' Sullivan

Lecturer in Mathematics Education,

School of Education,

National University of Ireland, Galway.

Written Reference 2

Galway Community College

091 755 464

23/04/21

To Whom it may concern,

Cian Mac Liatháin completed his third and final year teaching practices with us in GCC from Oct 2019-Dec 2020. Cian's teaching subject is Maths.

As PME coordinator for the school I was responsible for interviewing and placing Cian as well as supporting him along with his cooperating teacher Bernie Moran over the time he was here.

Cian was always punctual and thoroughly prepared for his classes. His attendance was excellent throughout his placement. Cian was eager to take on extra duties and responsibilities and so we offered him after school study supervision. During this time, he not only supervised but also offered help to any students who needed it in Leaving Cert Maths.

Cian had a good rapport with his students and is very knowledgeable in his subject. He has a very gentle nature and a positive attitude, and I have no doubt he will continue to influence students positively in the future.

If you have any questions regarding Cian, please feel free to contact me at the above address or by email at evelyn.jackson@gretb.ie.

Sincerely,

Evelyn Jackson

Apple Customer Satisfaction Elite Award Nomination

YOU & your customers

During his time at Apple Cian has delivered exceptional results out performing all of his colleagues. For example here are some of the results he has delivered for the last two quarters: CSAT 95.41%, IR 85.29%, AHT 14.17 & T2 Escalation rate 10.43%. Some of the best comments from his customers we can see below:

"Cian went over and above to help solve my issue. Would definitely want to speak with him again. He was patient and helped to solve my issue. Cian is an asset to your company and was a pleasure to speak with."

"Cian has resolved my problem with extreme efficiency, extreme patience and was very confident and reassuring from the very start."

"I can not praise Cian enough for the way that he handled my call with the patience and understanding that one normally only receives from Health Care professionals or fellow sufferers. On a scale of 1 to 10 I would rate Cian 11 so I therefore could not suggest anything better that could have been done to improve the service"

YOU & your team

Cian's input in the team is really good. He is an active member in the team chat and does his part in creating a positive atmosphere amongst the team. Cian contributes to team spirit being very positive and taking initiative in helping others.

"The advisor knew what he was doing and had the necessary groundwork completed prior to contacting Account and Billing"

"From my point of view Cian has been a pleasure to work with. He has shown great enthusiasm and professionalism from the outset. He has been eager to improve and actively searches for feedback in that regard. He is always reliable when I reach out to the team providing a quick response and support when he can lend it. Has been active also in helping us out in busy periods with Additional Hours where his schedule will permit it. He is a nice genuine guy to speak with."

"Emails with best practices and tips to the group is another area in which Cian has shown his interest in helping the team to perform well - increasing awareness of workarounds for various technical issues"

YOU & your manager

Apple takes great pride in providing stellar service to our customers in each and every interaction. Indeed, our business model has been built on the very principal of surprising and delighting and enriching our customer lives. It is no coincidence that we continue to top opinion polls for providing the best customer service and add countless awards to our trophy cabinet for being the best in class!

Quality and enhanced service delivery is a pivotal component of Apple's strategy. It is inherently linked to customer loyalty and successful growth.

"Any definition of good service is incomplete if not based on customer views" (Terry, 2012).

Based exclusively on our customer's opinions, it is with immense pride that we are delighted to welcome Cian Lyons again to the Apple Hall of Fame as an AppleCare CSAT Elite nominee!

Cian's outstanding commitment to provide best support to our Customers as well as contribution to his AppleCare Colleagues has warranted a place among CSAT Elite Nomination. Cian, keep up the excellent work you have produced so far in Apple!